The Goal: Stopping Small Mistakes from Becoming Big Disasters

In a medical company, if a lab error or a warehouse mistake isn't fixed quickly, the government (FDA) could shut the company down. This project was designed to make sure the most dangerous mistakes are caught and fixed immediately.

Step 1: Organizing the Warehouse (Data Engineering)

- **What you did**: You created a digital "filing cabinet" (a database) to hold all the company’s incident records.

- **Why you did it**: Information is often scattered across different emails and spreadsheets. By putting it all in one organized place, you ensured that no record could be lost or accidentally deleted.

Step 2: Identifying the "Fires" (Risk Analytics)

- **What you did**: You taught the system how to tell the difference between a minor typo and a major safety risk.

- **Why you did it**: Managers are too busy to read hundreds of reports every day. You programmed the "brain" of your system to ignore the small stuff and only flag the **2 most dangerous issues** that needed a boss’s attention right away.

Step 3: Setting the "Emergency Alarm" (Automation)

- **What you did**: You built an "Automatic Alert" system.

- **Why you did it**: Dashboards are great, but someone has to remember to log in to see them. Your system doesn't wait; it scans the records every hour and sends an **instant, urgent email** to the right manager the second a high-risk mistake is found. This ensures the "fire" is put out before it spreads.

Step 4: The Command Center (Visual Dashboards)

- **What you did**: You created two "Control Panels" (in Tableau and Power BI) that use big, red numbers to show the current risk level.

- **Why you did it**: You wanted the Big Boss (the CEO or Director) to be able to look at one screen for five seconds and know exactly which departments (like the Lab or Manufacturing) are having the most trouble.

Step 5: The Double-Check (Technical Audit)

- **What you did**: You used a specialized "Auditor" tool (Python/Google Colab) to double-check the math and the logic.

- **Why you did it**: In the medical world, accuracy is everything. This final step proved that your automated alarm was 100% accurate and that the information shown on the dashboards was the absolute truth.

The "Big Win" for the Company

Because of your work, the company moved from being **reactive** (finding out about problems weeks later) to being **proactive** (knowing about a problem the minute it happens).